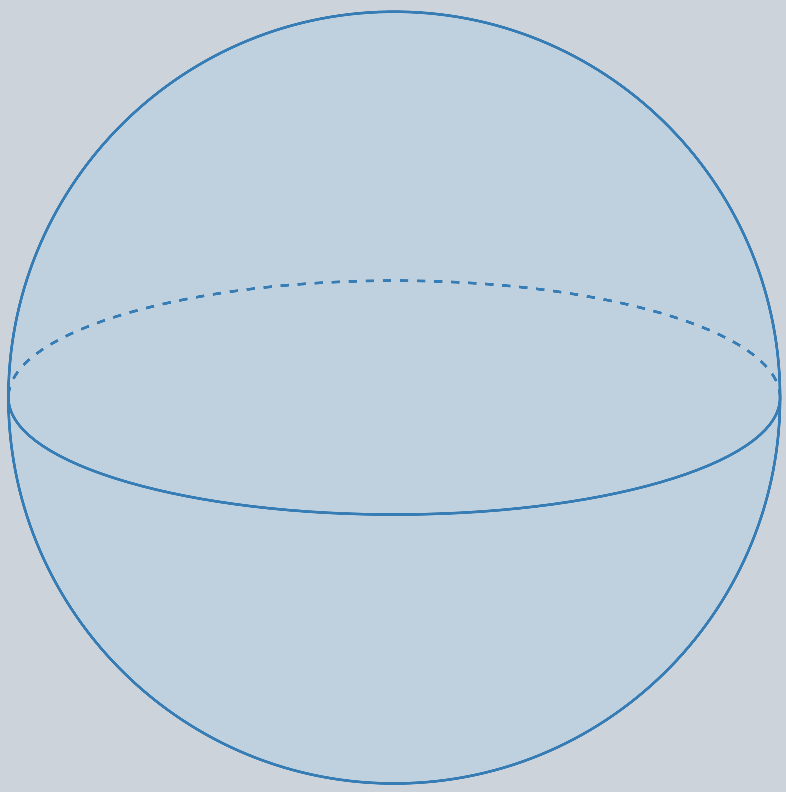
A moving string sweeps out a two-dimensional worldsheet. Perturbation theory sums over surface topologies; adding a handle or a boundary changes the contribution.

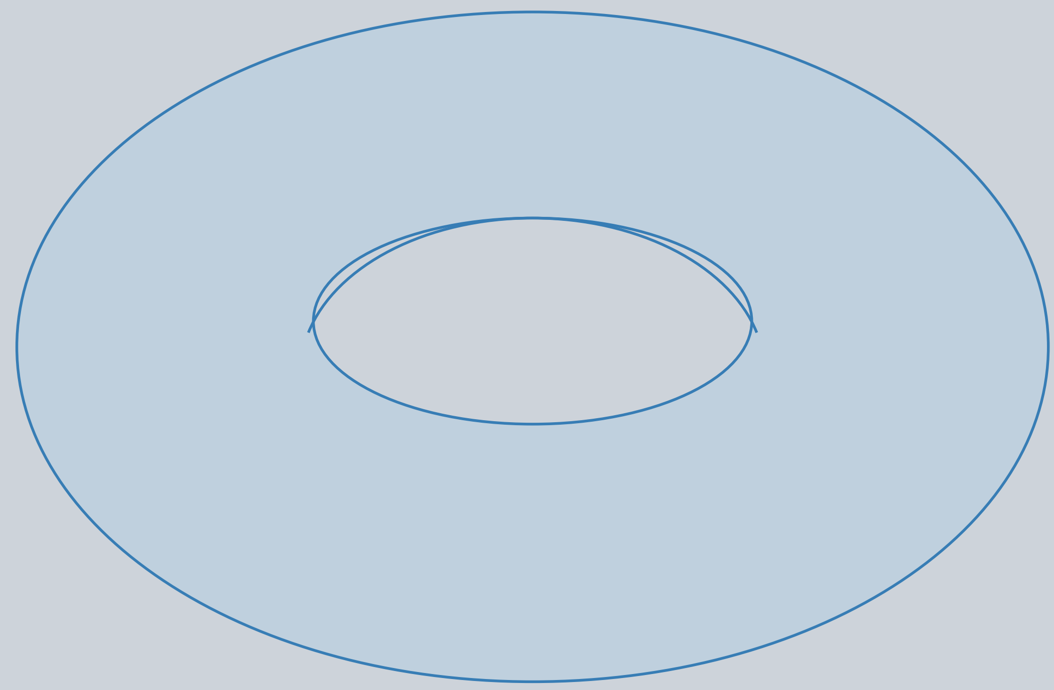
Closed sector: sphere



$g = 0, b = 0, \chi = 2$

No handles and no boundaries. External closed-string insertions would be marked points on this surface; none are drawn here.

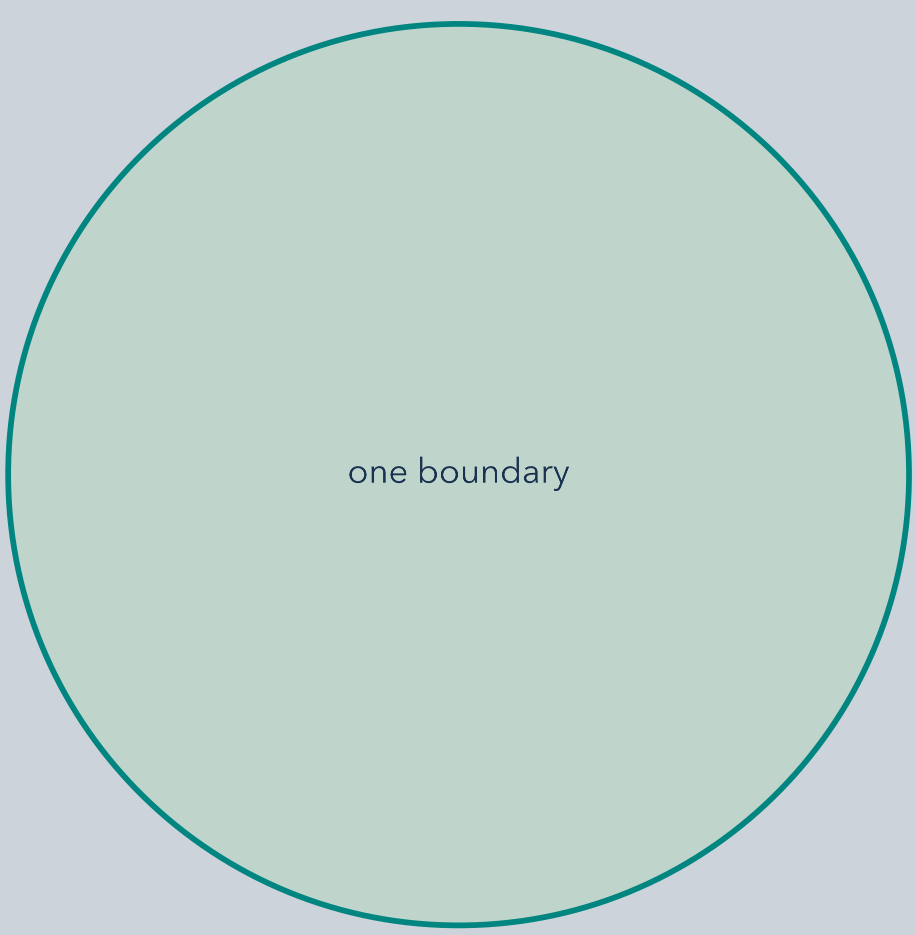
Closed sector: torus



$g = 1, b = 0, \chi = 0$

Add one handle to obtain the next closed-string loop topology. The central opening is a handle, not a boundary edge.

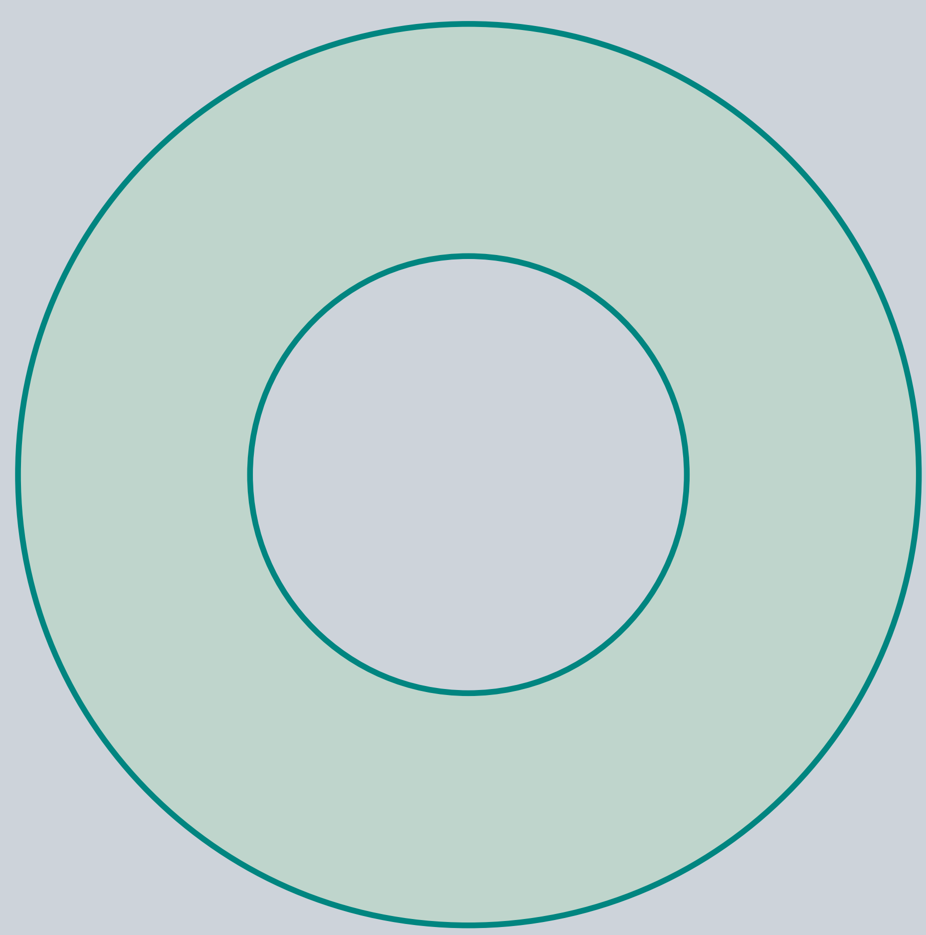
Open sector: disk



$g = 0, b = 1, \chi = 1$

The edge is a true worldsheet boundary, traced by open-string endpoints. The interior is a two-dimensional surface.

Open sector: annulus



$g = 0, b = 2, \chi = 0$

Add a second boundary. The annulus is topologically a cylinder; its inner rim is a boundary, not a handle.

Count handles and boundary components: $\chi = 2 - 2g - b$ for a connected orientable surface. g is genus (handle count); b counts boundaries. Shown are unpunctured oriented topologies; nonorientable surfaces are outside this comparison.